

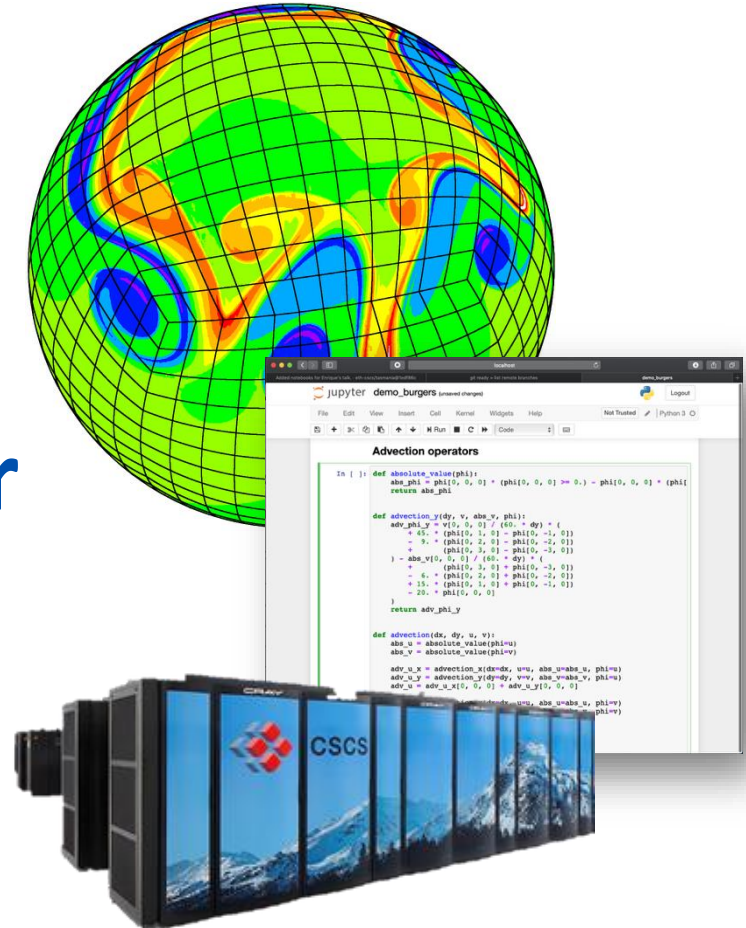
High Performance Computing for Weather and Climate (HPC4WC)

Content: Graphics Processing Units

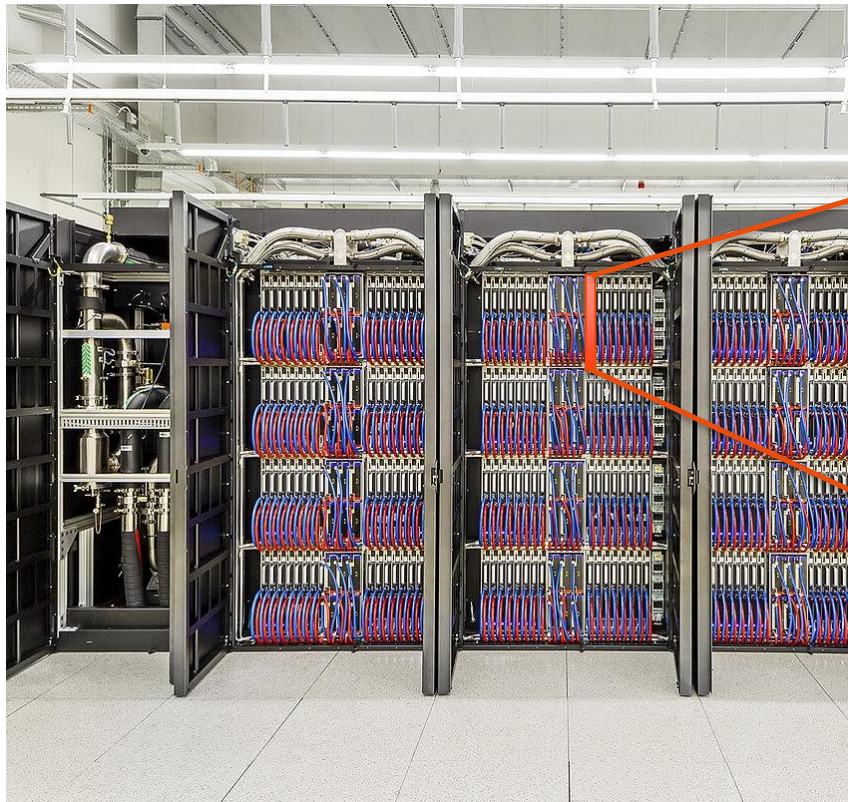
Lecturer: Simon Adamov, Oliver Fuhrer

Block course 701-1270-00L

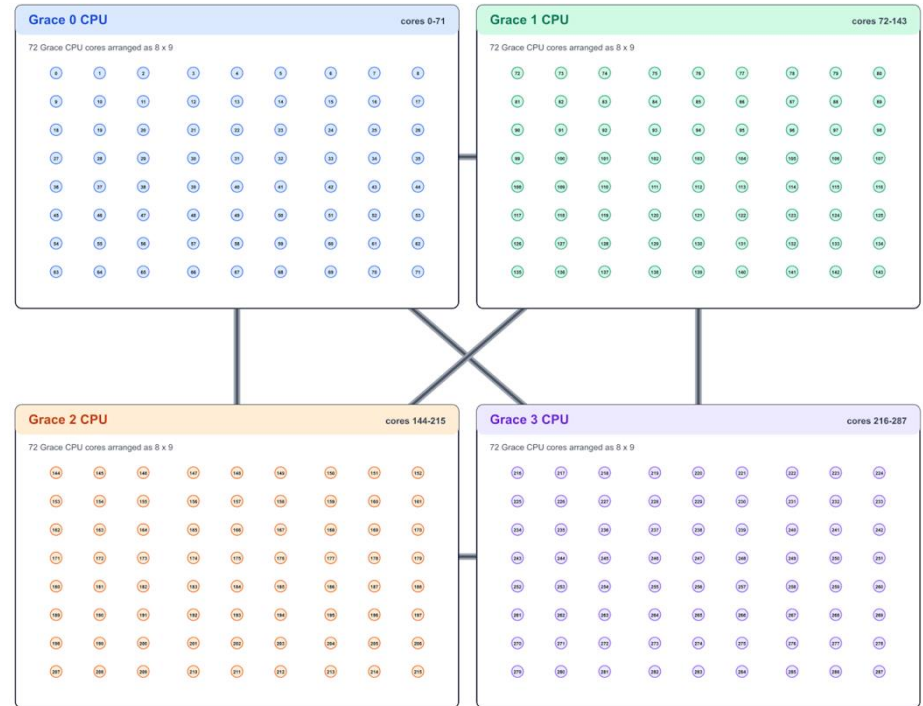
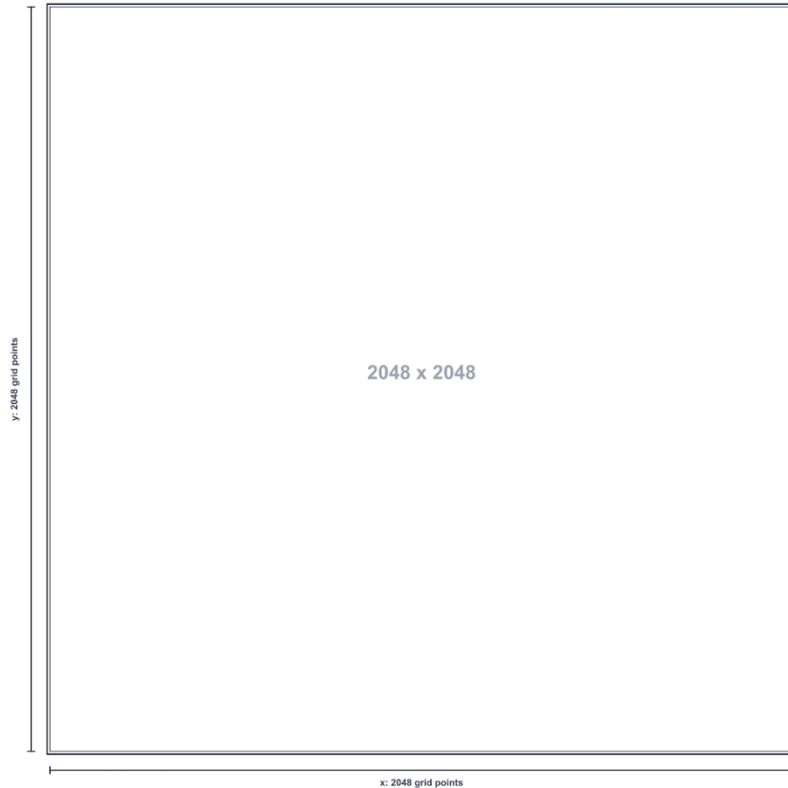
Summer 2026



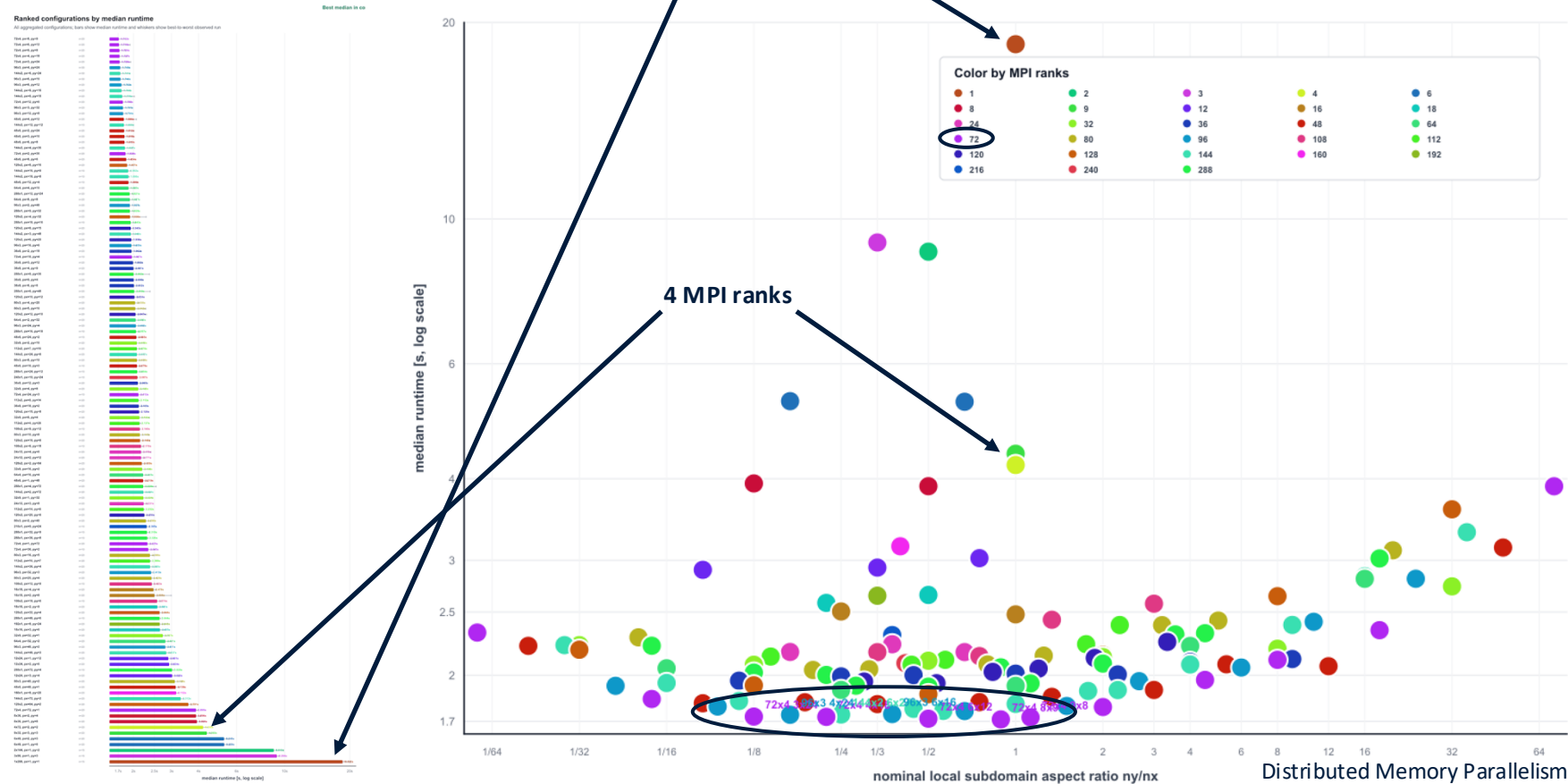
How does a node look like on Alps?



How to map this onto an Alps node?



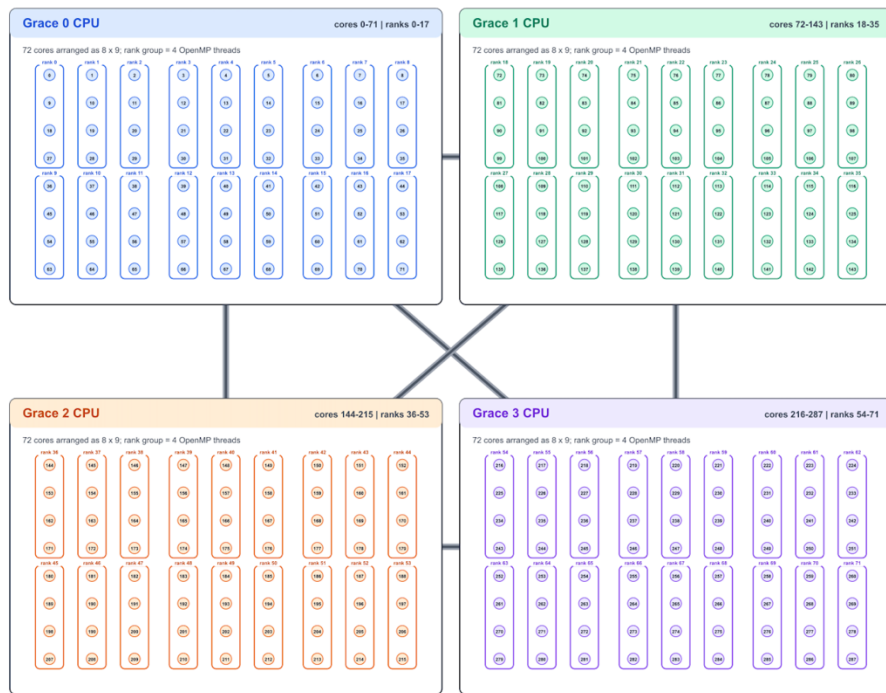
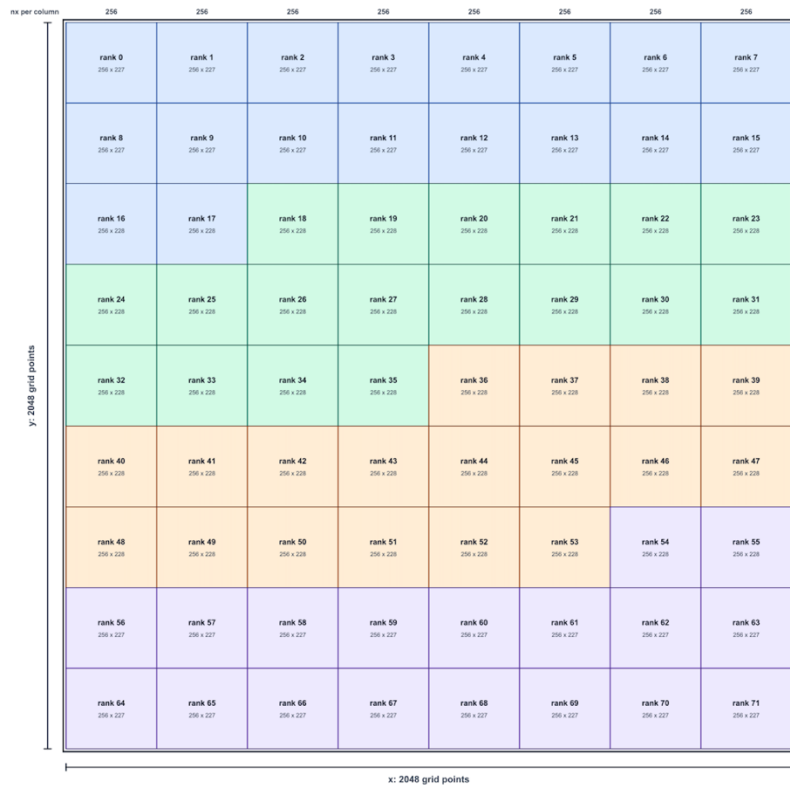
Choices matter!



The winning configurations

config	% cores	px	py	aspect	best	median	mean
72 x 4	100%	8	9	0.89	1.637	1.712	1.731
72 x 4	100%	6	12	0.50	1.652	1.716	1.744
72 x 4	100%	9	8	1.12	1.640	1.724	1.740
72 x 4	100%	4	18	0.22	1.654	1.727	1.736
72 x 4	100%	3	24	0.12	1.690	1.729	1.747
96 x 3	100%	4	24	0.17	1.633	1.740	1.746
144 x 2	100%	6	24	0.25	1.645	1.741	1.738
96 x 3	100%	6	16	0.38	1.599	1.744	1.744
96 x 3	100%	8	12	0.67	1.640	1.763	1.777
144 x 2	100%	9	16	0.56	1.679	1.764	1.779
144 x 2	100%	8	18	0.44	1.640	1.776	1.773
72 x 4	100%	12	6	2.00	1.750	1.788	1.800
96 x 3	100%	3	32	0.09	1.685	1.791	1.777
96 x 3	100%	12	8	1.50	1.720	1.794	1.796

The winning node mapping

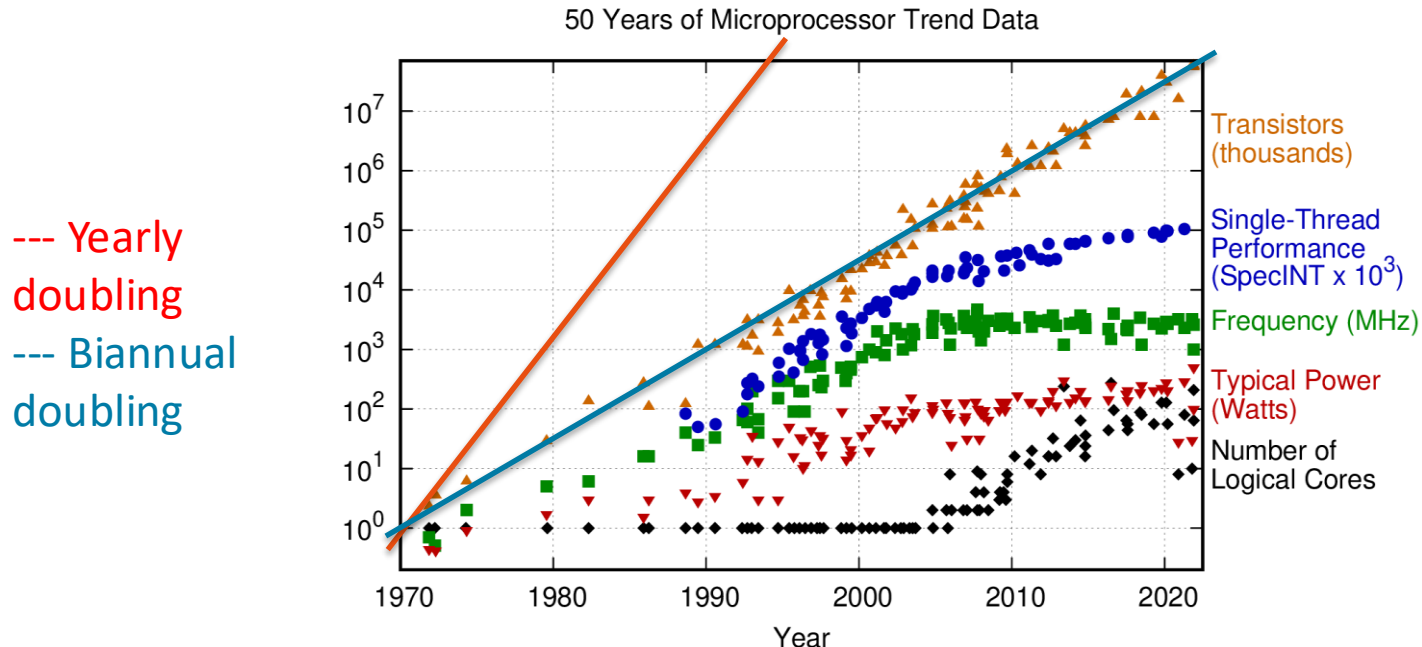


Learning Goals

- Understand why specialized hardware such as GPUs has become the new norm
- Learn how to program a GPU using a high-level programming language
- Grasp the potential and difficulties of GPU computing

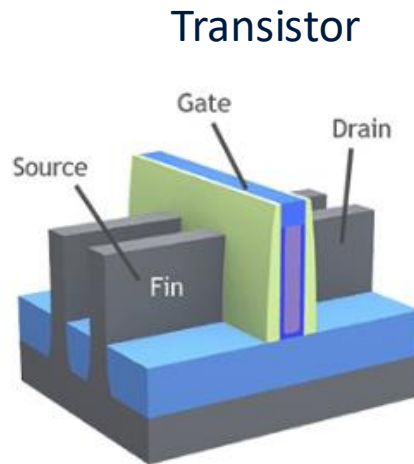
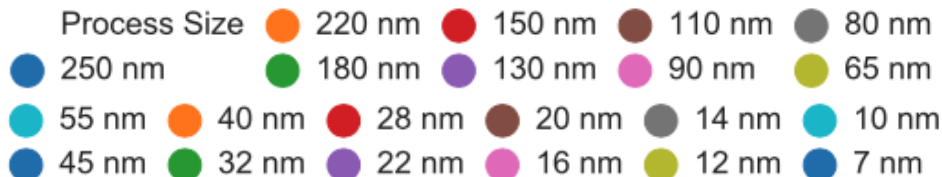
Moore's "Law" (1965)

- "The number of transistors in a dense integrated circuit will double every two years"



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2021 by K. Rupp

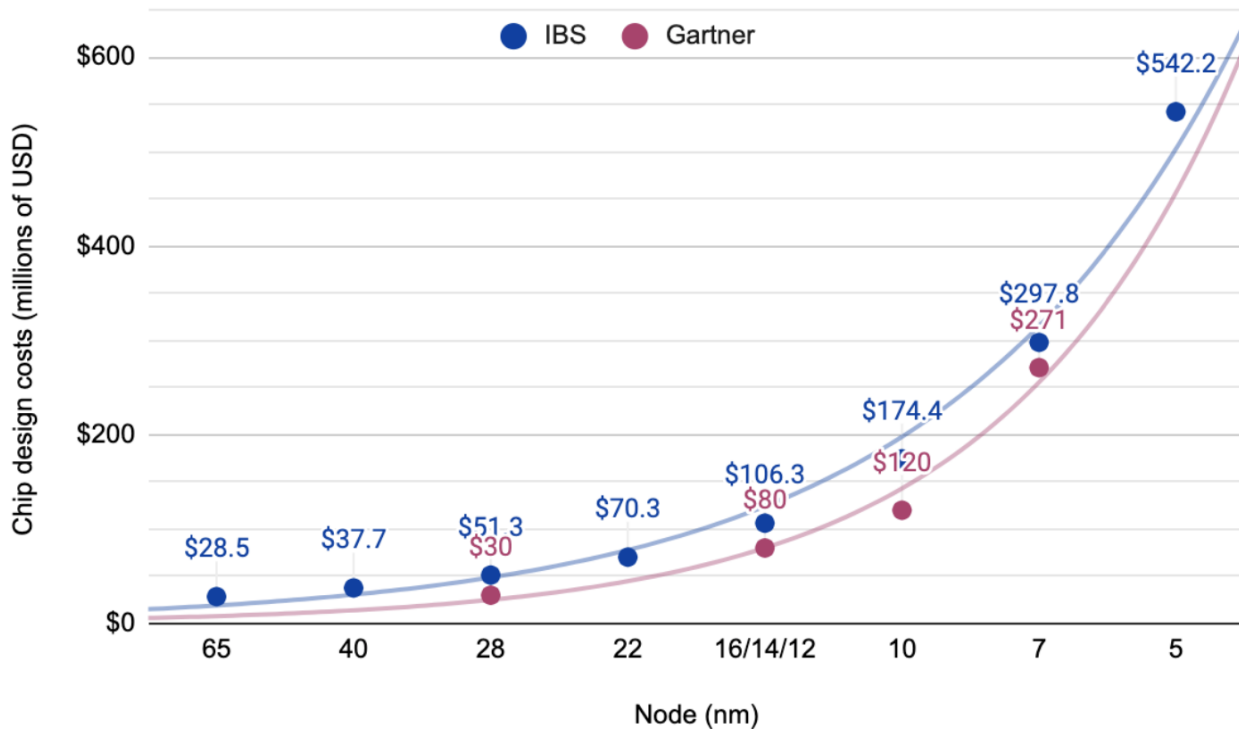
The End of General Purpose Computing?



Process size used to be a physical length scale.
Today it is mainly a technology-generation label.

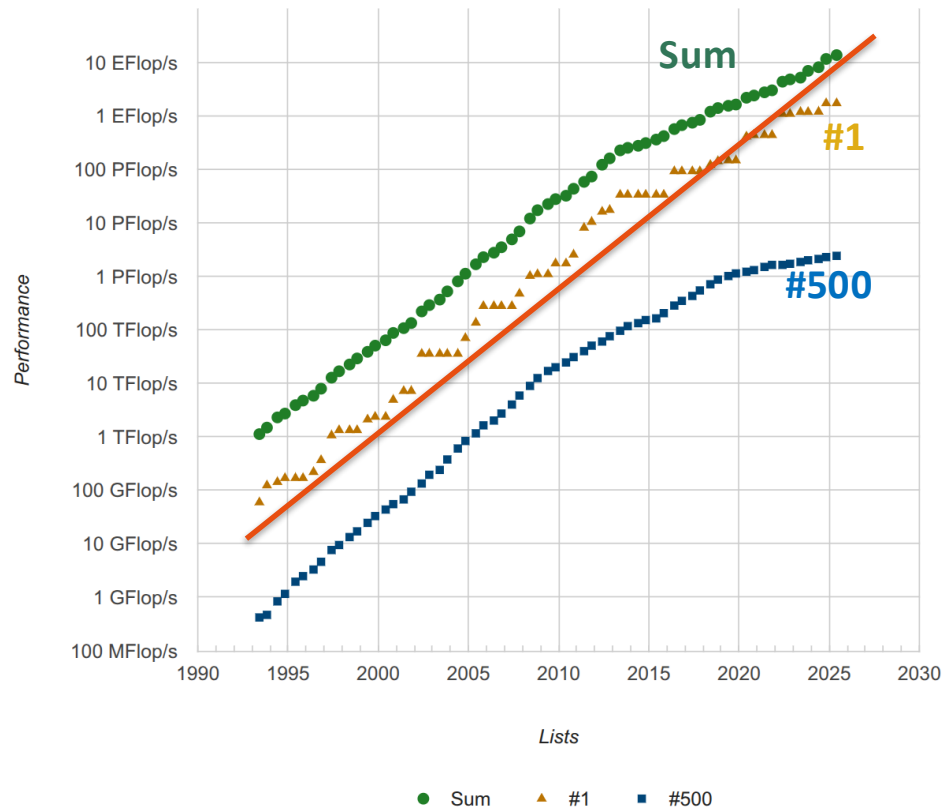
A smaller node usually means denser and more energy-efficient chips, but “2 nm” does not mean that the transistor is 2 nm large.

Chip Design Costs Increase



How Do the Most Powerful Systems Perform?

Performance Development



--- Yearly
doubling

So why are we (still) ok?

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	LineShine - LingKun, LX2 304C 1.55GHz, LingQi, Kylin OS, Shenzhen Cloud Computing Center Co., Ltd. National Supercomputing Centre in Shenzhen (NSCS) China	13,789,440	2,198.40	2,735.82	42,220
2	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
3	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
4	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
5	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, Bull EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794

AMD GPU

AMD GPU

INTEL GPU

NVIDIA GPU

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
6	HPC7 - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, RHEL 9, HPE Eni S.p.A. Italy	3,461,472	571.50	861.13	8,735
7	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure United States	2,073,600	561.20	846.84	
8	HPC6 - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461
9	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
10	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	574.84	7,124

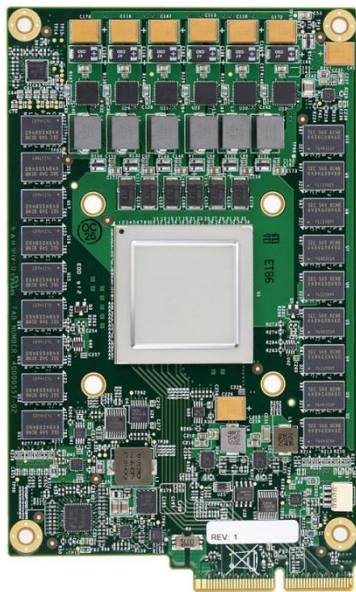
AMD GPU

NVIDIA GPU

AMD GPU

NVIDIA GPU

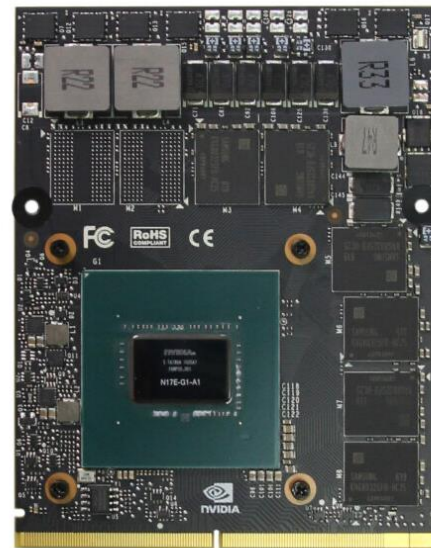
Specialized Chips are on the Rise!



Google's TPU
(e.g. machine learning)

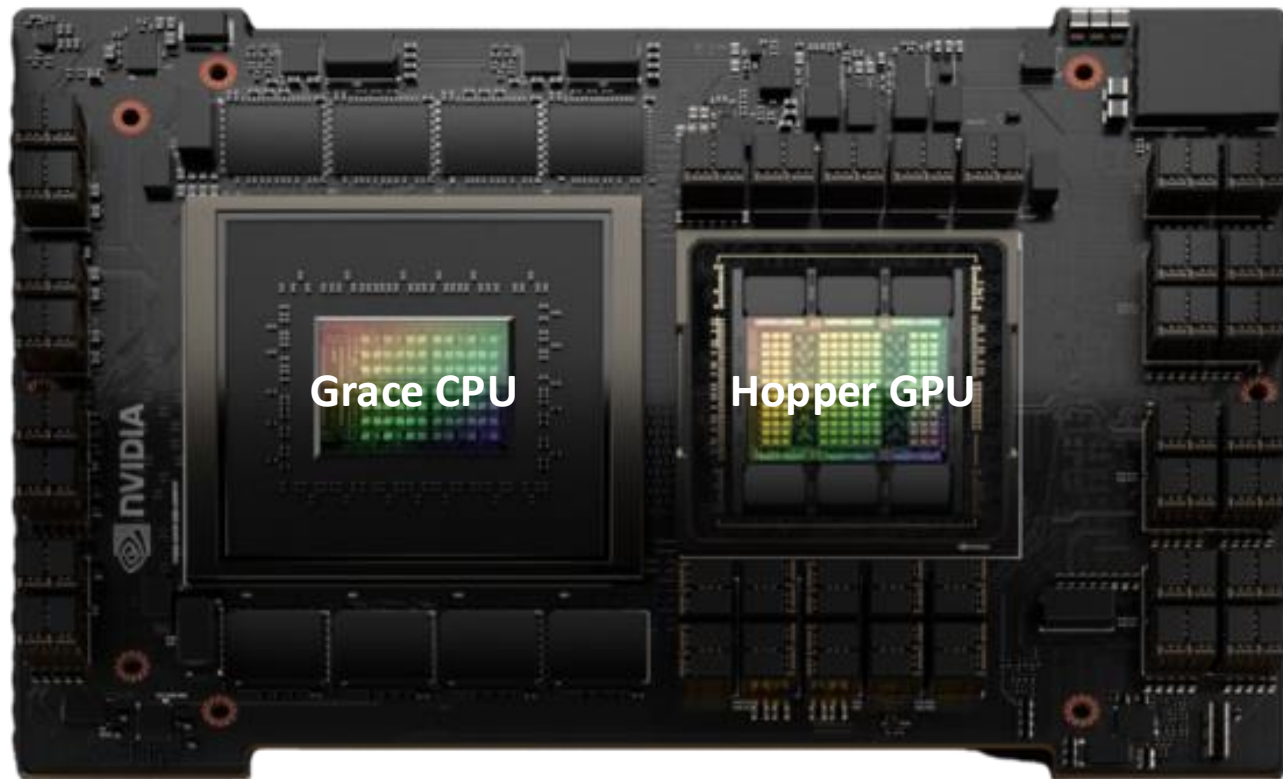


FPGA
(e.g. bitcoin mining)



GPU
(e.g. AI, gaming)

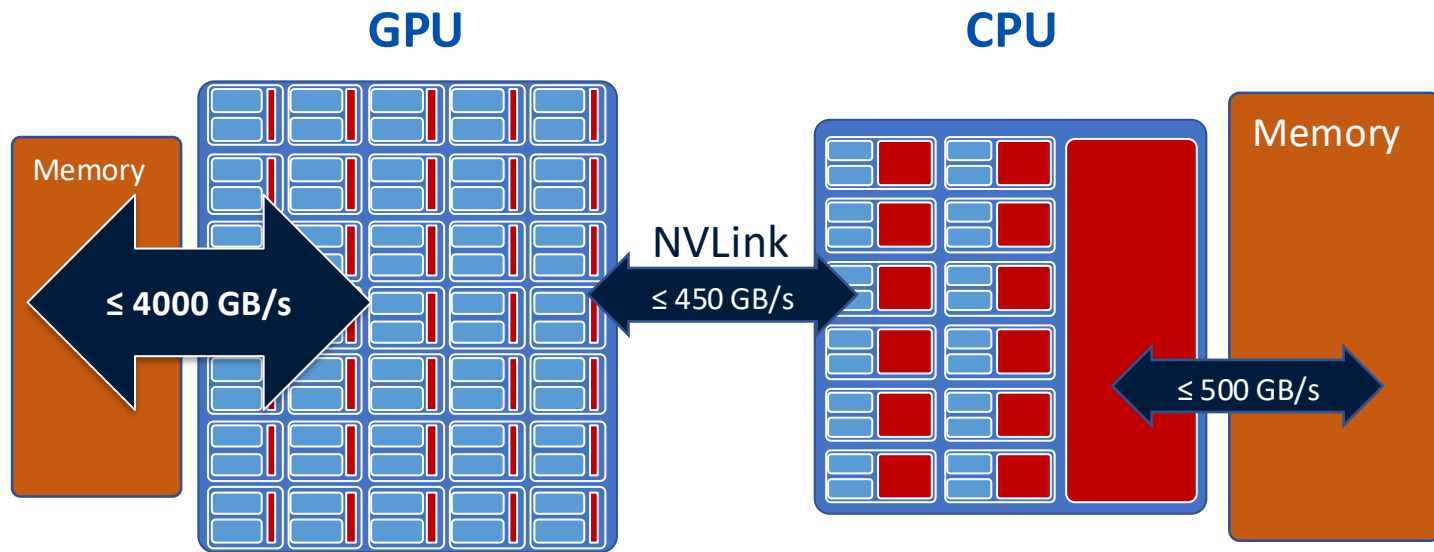
The NVIDIA GraceHopper 200



~ 200 W ≤ 2.5 TFLOP/s ≤ 500 GB/s

~ 700 W ≤ 34 TFLOP/s ≤ 4000 GB/s

Node Architecture



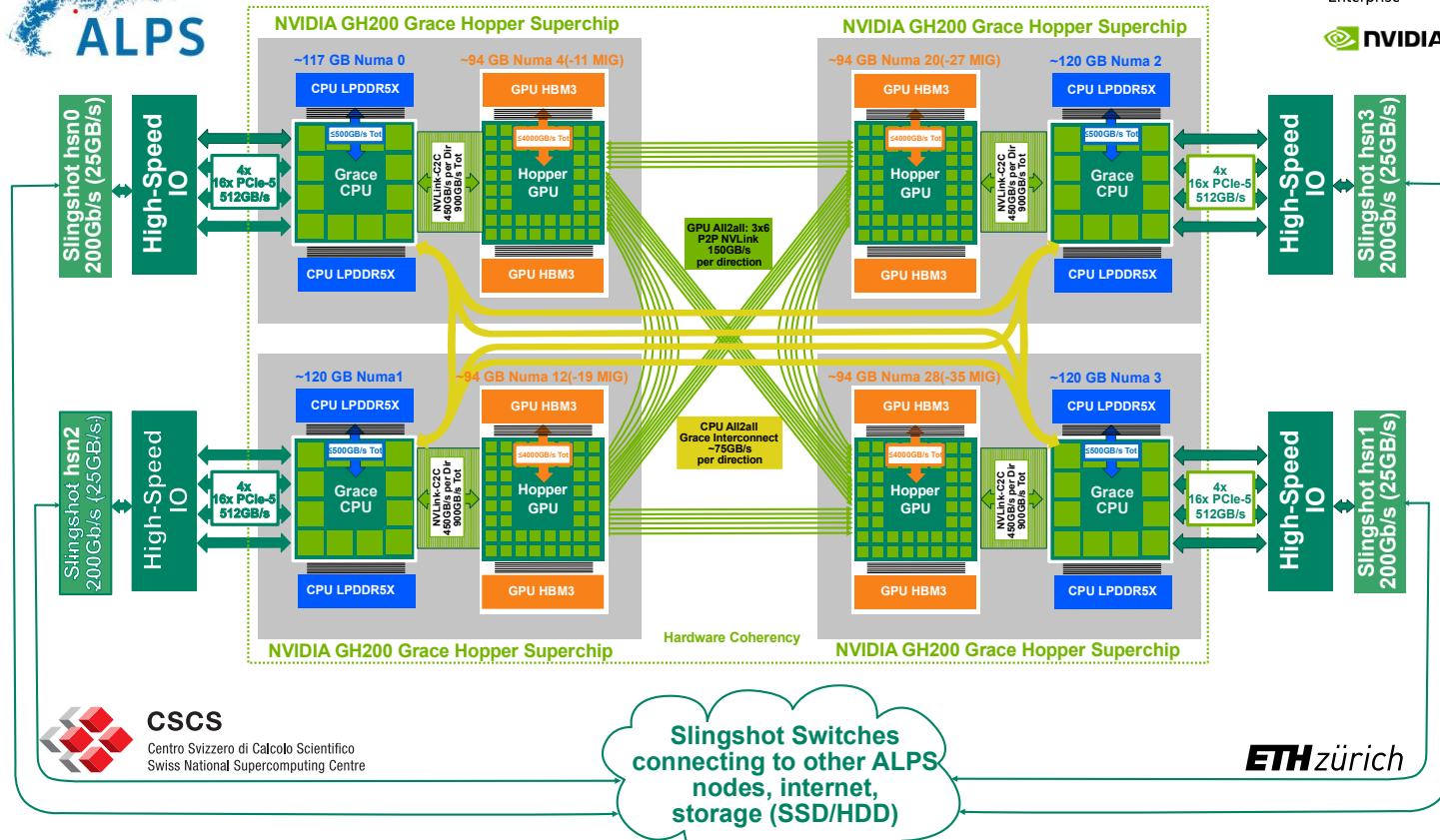
Important to minimize memory transfers between CPU and GPU!

Node Architecture

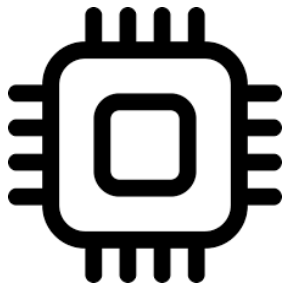


CSCS ALPS HPE Cray EX GH200 Node

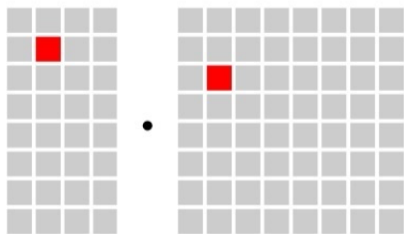
4xGH200 NVIDIA Grace Hopper Superchips ~477 GB (CPU) + ~378 GB (GPU) = ~855 GB Unified memory



CPU vs. GPU

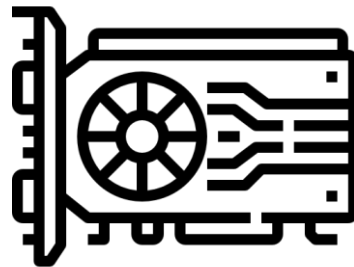


Latency



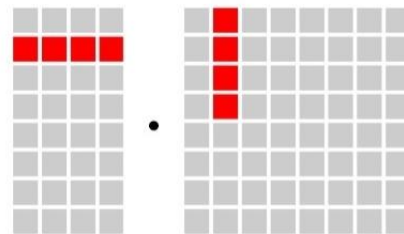
scalar

Architecture



Optimization

Compute Primitive



vector

Bandwidth

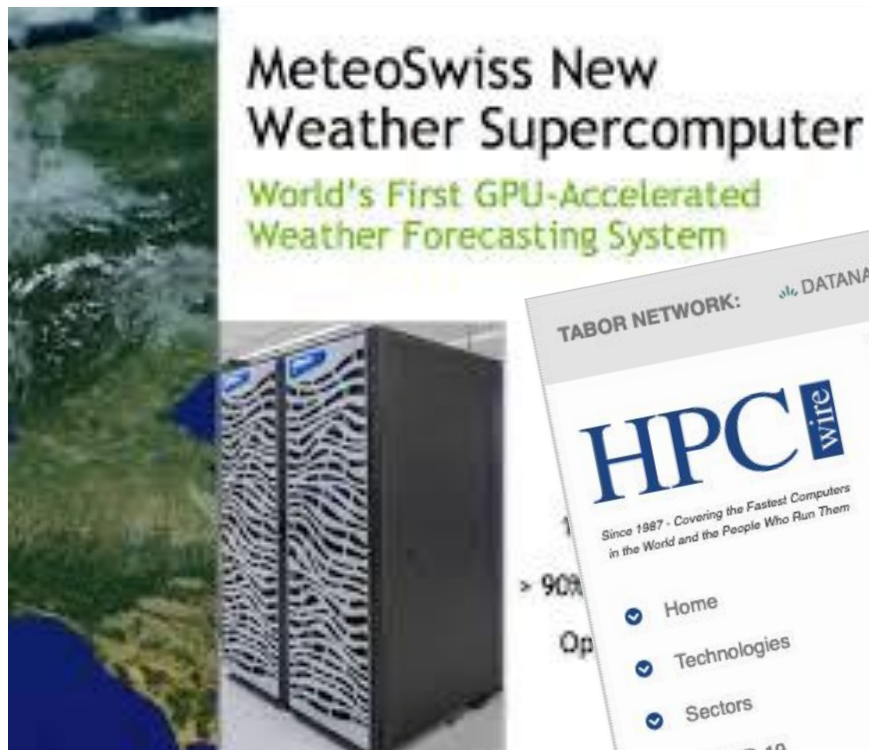
Hybrid Supercomputer - ALPS



~ 90% BW

~ 90% FLOP/s

Weather and Climate on GPUs



Programming GPUs

Libraries



Directives



Programming Models



Lab Exercises

01-GPU-programming-cupy.ipynb

- Introduction to GPU programming using a high-level programming language